

PATHPULSE

AI-Powered Road Infrastructure Intelligence

A Computer Vision & Exasol Analytics Pipeline

Team MindSpark 🎓 (1st-Year Students)
Exasol AI Build Challenge 2026

1. The Problem: Blind Spots in Civic Infrastructure

Manual & Slow

City governments currently rely on manual citizen reports or slow, expensive physical surveying fleets to map out road damages.

Reactive, Not Proactive

By the time a pothole is officially reported, it has already caused severe vehicle damage, accidents, or massive traffic congestion.

The Data Silo Dilemma

Incident reports are scattered across emails, phone calls, and outdated databases. It is impossible for city planners to analyze city-wide infrastructure trends or deploy resources effectively without a centralized analytics engine.

2. The Solution: PathPulse

PathPulse is an **end-to-end computer vision and analytics pipeline** that automates road safety monitoring. By mounting a camera on a city vehicle, we turn raw video feeds into actionable infrastructure intelligence.

- **Data Source:** Trained on the **RAD (Road Anomaly Detection) Dataset** from Kaggle to accurately detect real-world infrastructure hazards.
- **Automated Detection:** Feed dashcam or traffic camera footage directly into our AI to instantly identify road hazards as the vehicle drives.
- **Actionable Intelligence:** We categorize anomalies, score their severity, and log them directly into an enterprise analytics database.
- **Live Command Center:** A real-time dashboard built for city planners to see exactly where the infrastructure is failing across the entire grid.

3. The Tech Stack: An Enterprise-Grade Pipeline

The Brain (YOLOv8)

We trained a custom Ultralytics YOLOv8 Nano computer vision model to detect 6 specific classes: Heavy Vehicles, Light Vehicles, Pedestrians, Road Damages, Speed Bumps, and Unsurfaced Roads.

The Memory (Exasol)

The AI's telemetry is streamed live into a local Dockerized **Exasol Analytics Database**. Exasol was chosen specifically for its ultra-fast, in-memory analytical processing capabilities.

The Face (Streamlit)

An interactive Python frontend that allows operators to dynamically adjust AI Non-Max Suppression (IoU) and Confidence thresholds while viewing live, zero-latency database metrics.

4. Overcoming Hackathon Hurdles

- **Data Balancing:** The original Kaggle dataset was heavily skewed toward standard cars. We wrote a Python script to dynamically downsample the data, preventing the AI from becoming biased and forcing it to learn rare anomalies like potholes.
- **Speed vs. Accuracy:** We utilized YOLOv8 Nano and trained it for exactly 40 epochs—finding the perfect sweet spot between lightweight, real-time inference speed (30 FPS) and detection accuracy.
- **Database Bypassing:** We bypassed standard ORM bottlenecks by writing raw, parameterized batch-insertion SQL queries via the `pyexasol` websocket driver directly to the Exasol backend.
- **Edge-Computing Architecture:** As 1st-year students, we deployed Exasol locally via Docker. This mimics a true "edge-computing" environment (like a garbage truck's onboard computer), ensuring zero-latency data processing without internet connectivity!

5. Real-World Impact & Future Roadmap

Smarter Cities Now

Cost Reduction: Eliminates the need for manual road surveying fleets.

Resource Allocation: Exasol's analytics allow city planners to deploy repair crews based on data-driven severity heatmaps rather than guessing.

Future Roadmap

Geospatial Mapping: Integrating GPS coordinates to plot potholes on a live map.

Cloud Migration: Moving our local Exasol container to AWS for centralized data syncing.

Automated Dispatch: Triggering APIs to dispatch crews when a pothole severity hits Level 5.